

# Full Stack Development and Deployment of a Web-Based Speech Recognition System

**Weeks 35-36**  
**7 September – 20 September 2026**

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# Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
  - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
  - Improve robustness, usability, and AI-driven recording analysis



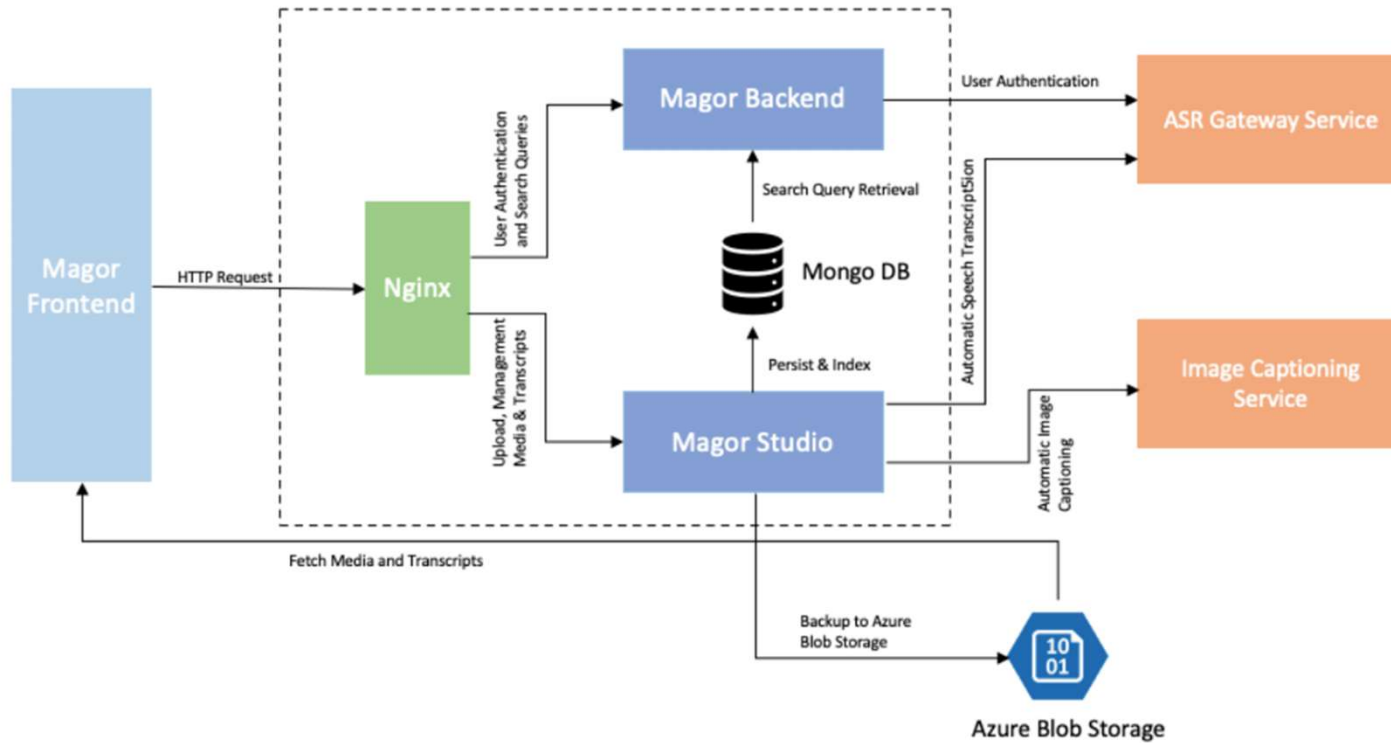
# Project Objectives

**The primary objectives of this project are:**

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



# Systems Overview



Source: Huang (2021)



# Agenda

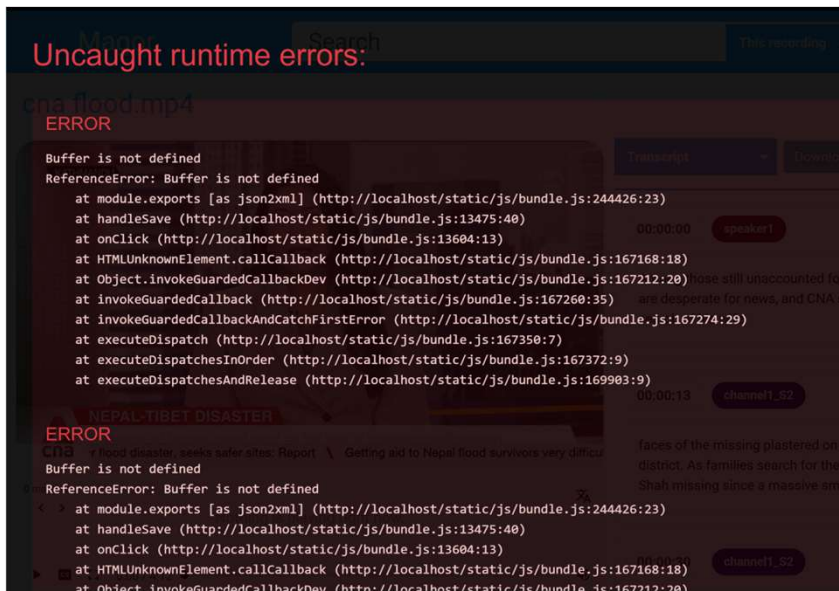
## Work Done Over The Weeks:

- Resolved XML dependency issues
- Fixed transcript selection in “Show Options” panel
- Implemented change transcript in transcript editor page
- Upgraded Studio/Backend Node Version and resolved dependency issues

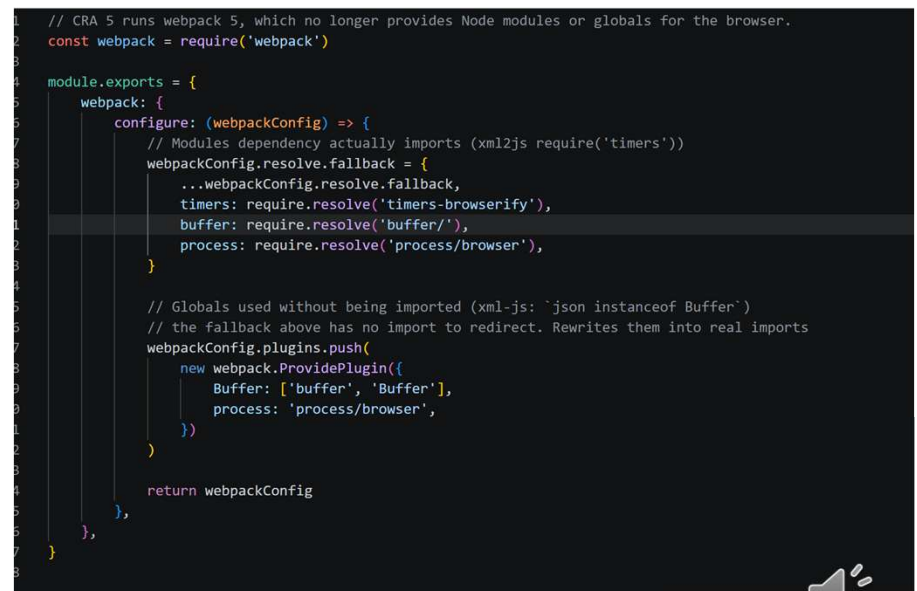


# Task 1 (Resolved XML Dependency Issue)

- XML libraries error:



Error when editing transcript



CRA workaround to revert behavior

# Task 1 (Resolved XML Dependency Issue)

- Solution – Use fast-xml-parser:
  - Old XML libraries (xml2js, xml-js) were **no longer maintained** and had reached their final releases
  - Replaced them with **fast-xml-parser**, which is actively maintained and browser-compatible, allowing CRACO to be removed
  - Created a **shared XML parser file** with reusable **parseXml** and **buildXml** functions

```
C:\> FYP > FYP > Code > current copy > magor-root > magor-frontend > src > Magor > Utils > JS xmlParser.js
1  import { XMLParser, XMLBuilder, XMLValidator } from 'fast-xml-parser'
2  /*
3   * Shared XML handling for transcripts
4   * Used by transcriptProvider, Recording/index and TranscriptEditor
5   */
6
7  // ensure nested elements are always parsed as arrays, even if there's only one
8  const isArray = (name, jpath, isLeafNode, isAttribute) =>
9    !isAttribute && jpath.includes('.')
10
11  const options = {
12    ignoreAttributes: false,
13    attributesGroupName: '$',
14    attributeNamePrefix: '',
15    textNodeName: '_',
16    ignoreDeclaration: true,
17    isArray,
18  }
19
20  const XML_DECLARATION = '<?xml version="1.0" encoding="UTF-8" standalone="yes"?>'
21
22  export const parseXml = (text, { processAttributeValues = true } = {}) => {
23    const validation = XMLValidator.validate(text)
24    if (validation !== true) {
25      throw new Error(`Invalid XML: ${validation.err.msg}`)
26    }
27    return new XMLParser({
28      ...options,
29      parseAttributeValue: processAttributeValues,
30      parseTagValue: processAttributeValues,
31    }).parse(text)
32  }
33
34  export const buildXml = (object) => {
35    const xml = new XMLBuilder({
36      ...options,
37      format: true,
38      indentBy: ' ',
```



## Task 2 (Fixed Show Options Panel Change Transcript)

- Show Options Change Transcript Issue:

The screenshot displays the Magor video player interface. At the top, there is a blue header bar with the text "Magor" on the left, a search bar in the center, and "This recording" with a magnifying glass icon on the right. Below the header, the video title "fyp faulty video.mp4" is shown, followed by the subtitle "Video to investigate". A blue "Edit" button is located to the right of the title.

The video player itself shows a scene with a man in a black shirt and cap, a woman in a green shirt, and a child in a blue shirt. The video is paused at 0:00 / 8:50. Below the video, there is a subtitle "来 从你开始" and a status bar indicating "Nothing is playing right now."

On the right side of the video player, there is a transcript panel. The panel has tabs for "Transcript", "Ask AI", and "Hide options". The "Transcript" tab is active. It shows a list of transcript entries for "Speaker 1". A dropdown menu is open, showing a list of timestamps: "Thu, 10 Sep 2026 12:30:38 GMT", "Thu, 10 Sep 2026 12:20:21 GMT", and "Thu, 10 Sep 2026 12:30:38 GMT". The third option is selected and highlighted in blue. Below the dropdown, there are checkboxes for "Show transcript", "Show Image Captions", and "Show Sound Captions". There are also tags for "fyp" and "faulty".

Below the transcript panel, there is a "Top Words" section with a list of words: "people", "oil", "time", "government", "open", "protest", "world", "country", "price", "demand", "today", "virus", "economy", "protesters", and "testing".

At the bottom right of the interface, there is a speaker icon.

## Task 2 (Fixed Show Options Panel Change Transcript)

```
<div className="ops selectTranscript optionsPanelTranscript">
  <label>Transcript:</label>
  <select
    className="optionsPanelSelect"
    value={transcript?.azureResourceUrl || ''}
    onChange={(e) => {
      const match = transcripts.find(
        (t) => t.azureResourceUrl === e.target.value
      )
      if (match) setTranscript(match)
    }}
  >
    {transcripts.map((t) => (
      <option key={t.azureResourceUrl} value={t.azureResourceUrl}>
        {t.name}
      </option>
    ))}
  </div>
```

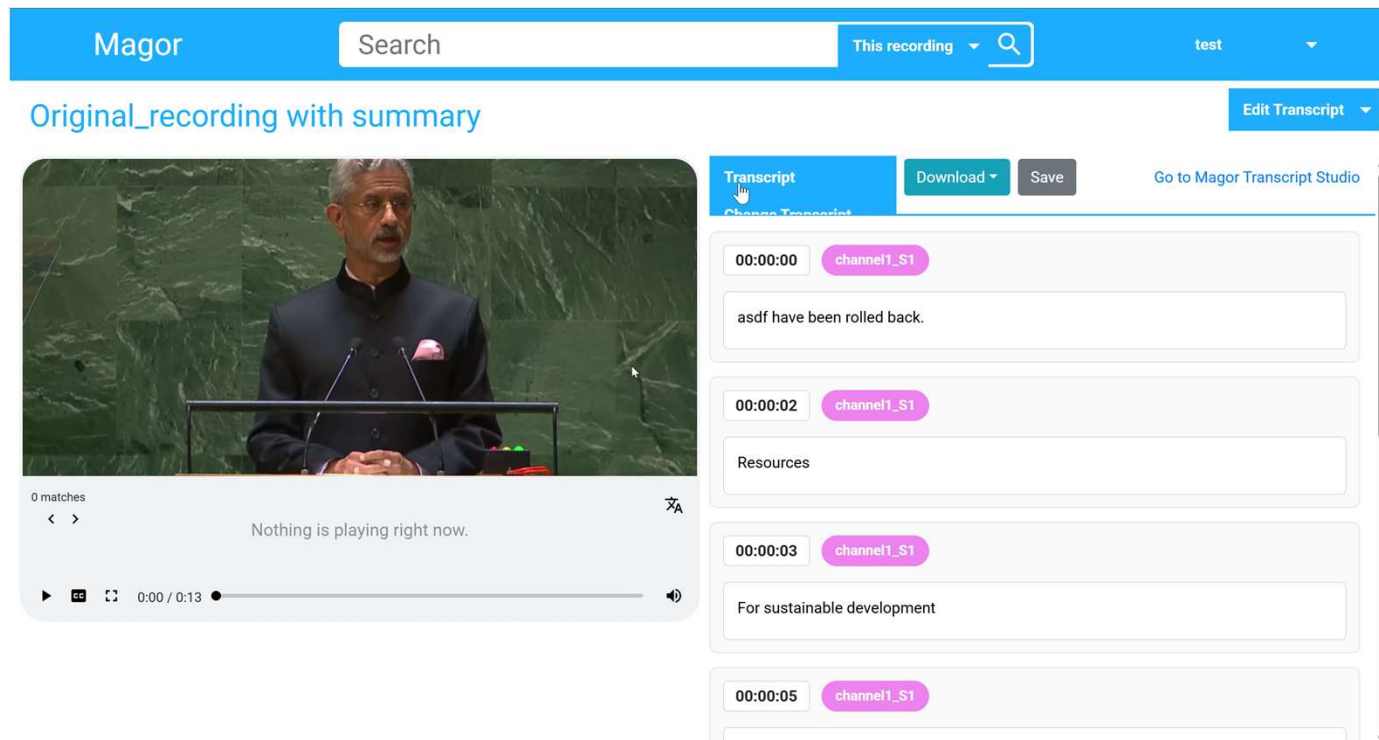


```
<div className="ops selectTranscript optionsPanelTranscript">
  <label>Transcript:</label>
  <select
    className="optionsPanelSelect"
    value={transcript?._id || ''}
    onChange={(e) => {
      const match = transcripts.find(
        (t) => t._id === e.target.value
      )
      if (match) setTranscript(match)
    }}
  >
    {transcripts.map((t) => (
      <option key={t._id} value={t._id}>
        {t.name}
      </option>
    ))}
  </div>
```



## Task 3 (Change Transcript in Transcript Editor)

- Editor Dropdown No Functionality Demo:



The screenshot displays the Magor Transcript Editor interface. At the top, a blue header bar contains the 'Magor' logo, a search bar, and a dropdown menu labeled 'This recording' with a magnifying glass icon. Below the header, the text 'Original\_recording with summary' is visible, along with an 'Edit Transcript' button. The main content area is divided into two sections. On the left is a video player showing a man in a dark suit speaking at a podium. The video player controls at the bottom indicate '0 matches' and 'Nothing is playing right now.' with a progress bar at 0:00 / 0:13. On the right is the 'Transcript' editor, which has a 'Transcript' tab selected. It includes 'Download' and 'Save' buttons, and a link to 'Go to Magor Transcript Studio'. The transcript list contains four entries, each with a timestamp and a channel label 'channel1\_S1':

- 00:00:00: asdf have been rolled back.
- 00:00:02: Resources
- 00:00:03: For sustainable development
- 00:00:05:



# Task 3 (Change Transcript in Transcript Editor)

- Fixed Transcript Editor Dropdown Demo:

The screenshot displays the Magor interface. At the top, there is a blue header bar with the 'Magor' logo, a search bar, and a dropdown menu set to 'This recording'. Below the header, the video title 'cna flood.mp4' is shown, along with a description: 'Hundreds of families await word on the fate of thousands still missing'. A red banner at the bottom of the video frame reads 'NEPAL-TIBET DISASTER'. The video player shows a news anchor in front of a cityscape. To the right of the video player, there is a transcript editor. It includes tabs for 'Transcript', 'Ask AI', and 'Show options'. The transcript content is as follows:

**Test1** test test Among those still unaccounted for are nearly 100 Chinese nationals whose relatives are desperate for news, and CNA spoke with one man who traveled to Kathmandu in search of his wife.

**Test2** faces of the missing plastered on walls at relief camps and hospitals in the disaster hit district. As families search for their loved ones, among them Chinese national Lily Shah missing since a massive smartlight struck the Nepal Tibet border a week ago.

**Test2** Her husband, 38 year old Guan Do last spoke to her the night before disaster struck.

**channel1\_S3** 就是很家常的一些话. 但是第二天中午我得知这边友谊士流之后, 应该是中国时间的11:30左右. 我再跟他联系就没有任何回应了.

**Test2** Miss Lee was working on a highway project in Nepal near China's Jurong border checkpoint.

At the bottom right of the interface, there is a speaker icon.

## Task 4 (Backend/Studio Node24 Upgrades)

- Node 24 Upgrade – Studio & Backend

- Upgraded both to Node 24
- nodejieba even after updates was incompatible with Node 24
- Switched to Node 24 Trixie (supports prebuilt binaries)
- All three services now run on Node 24:

```
● PS C:\FYP\FYP\Code\current copy\ma  
ker exec $s node -v)" }  
frontend-client: v24.20.0  
studio-server: v24.20.0  
backend-server: v24.20.0
```



# Plans for Next Update

- Improve Ask AI Summary and Q&A requests
  - Update Summary requests to use taskType: GENERAL and queue: meadow9
  - Update Q&A requests to use taskType: CHAT and queue: meadow9
  - Pass previous Q&A through the dedicated chatHistory attribute instead of embedding it in the prompt

